

64th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Immunomodulatory Microbial Metabolites Identified in Allo-SCT Patients Protect from GvHD By Sting Dependent Modulation of Intestinal Tissue Homeostasis

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Abstract Background: Graft-versus-host disease (GVHD) is a dreaded complication after allogeneic stem-cell transplantation (ASCT). Previously, we and others showed that activation of type I interferon (IFN-I) inducing pathways such as RIG-I/MAVS or cGAS/STING can promote the integrity of the intestinal barrier and limit GVHD. However, the signals that drive these protective IFN-I responses are poorly understood.

Commensal microbiota can (i) have distant effects on immune responses through modulation of IFN-I signaling and (ii) predict mortality in ASCT patients. We hypothesized that microbiota-derived products such as microbial metabolites engage IFN-I signaling in immune and non-immune cells poising them for induction of protective responses. We established a prospective, multi-centric clinical study in patients newly diagnosed with acute leukemia and performed longitudinal stool sampling to track changes in microbial community composition and metabolites expression levels (Submitted as a separate abstract to ASH 2022 (Orberg & Meedt et al.)). We showed that patients with high metabolite expression are less likely to develop GVHD. In this study, we translated our clinical observations to mouse models of acute GVHD and human and mouse intestinal organoid models to uncover the molecular mechanisms of protective metabolites during ASCT.

Methods: Stool samples from ASCT patients were obtained in Munich and Regensburg in accordance to IRB-approved study protocols. Patients were sampled at initial diagnosis (Dx), prior to conditioning and weekly after ASCT up to day 28. We analyzed samples by 16S rRNA sequencing and mass spectrometry. Next, we tested metabolites detected in patients *in vitro* in murine and human intestinal crypt-derived organoids and organoid/T cell coculture assays. To obtain a mechanistic understanding, we stimulated WT or IFN-I-signaling-impaired organoids or used small molecule inhibitors and analysed them regarding growth performance, cell composition and by scRNAseq. Finally, Metabolites were applied as treatment in pre-clinical intestinal damage models and GVHD mouse models. Outcomes were assessed by a novel *organoid recovery assay* in addition to established read-outs.

Results: Here, we present expression profiles of metabolites including short-chain fatty acids, DAT, ICA and secondary bile acids. Following ASCT, and especially at the early time-points, metabolite expression declined drastically. We confirmed this trend in our multi-centric cohort of ASCT patients by comparing levels of DAT and ICA sampled at admission to the transplantation ward (Conditioning) versus at clinical diagnosis of GVHD: in patients with GVHD, metabolite levels were drastically reduced. Metabolite stimulation of mouse small intestinal organoids promoted organoid growth and required intact STING signaling. Additionally, organoids showed an increased stemness and resistance to T cell mediated damage. Both healthy donor and GvHD patient derived human colon organoids also responded to DAT and ICA, however the metabolite effect was lost when co-administered with the STING-inhibitor H151. Next, we prophylactically administered metabolites in a major mismatch mouse ASCT model. Metabolite-treated mice showed significantly better outcomes in

barrier integrity and organoid based intestinal regeneration assays. Additionally, metabolite treatment was associated with an increased regulatory T cell infiltration. These effects were abrogated in STING^{-/-} recipients, but not in IFN α R^{-/-} recipients. Finally, metabolite treatment was protective in major mismatch GVHD mouse models.

Conclusions: We identify that microbial-derived metabolites detected in patients can engage the STING pathway in humans and mice to confer resistance from immune damage. Thus, prophylactic administration of metabolite cocktails or bacterial consortia may reduce occurrence of GVHD in ASCT patients.

Disclosures Holler: *Maat-Pharma (Lyon France):* Membership on an entity's Board of Directors or advisory committees; *Pharmabiome (Zürich, CH):* Membership on an entity's Board of Directors or advisory committees.

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